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Rating deflation versus inflation: On procyclical credit ratings

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ABSTRACT

This article provides a theoretical analysis to reconcile the controversy between rating deflation versus inflation. In our model, the credit rating agency trades off between the current incomes paid by the issuer upon receiving a favorable rating and the future reputation costs. We show that both rating deflation and rating inflation can occur in equilibrium. Furthermore, credit ratings are procyclical since the probability of default is higher and thus the reputation costs are higher during recessions than during booms.

Nobody establishes a rating agency in order to help anybody.

[The Polish prime minister, Donald Tusk.[1]]

1. Introduction

The "Big Three" credit rating agencies (CRAs), Standard and Poor's, Moody's, and Fitch, have played a critical role in the capital markets by assessing and spreading information about default likelihoods and recovery rates of securities. The quality of credit ratings is essential for the effective operation of the financial market, and huge losses could arise if rating agencies fail to provide accurate and timely ratings. However, there exist two opposing arguments on how they have actually performed (Bae et al., 2010). The first, inspired by recent global financial crisis, is the rating inflation view: CRAs are accused of being too cozy with the companies and the financial products they rate and bearing a responsibility for the subprime crisis. The

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[1] "EU leaders blame the euro crisis on American credit rating agencies" by Daniel Hannan, The Telegraph, July 7th, 2011.

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second is the rating deflation view: CRAs are blamed of being too focused on a company's or a country's bottom lines, rating or downgrading them too harsh.

Is it possible that the CRAs engage in both rating inflation and deflation? Are their rating standards related to business cycles? If so, are they procyclical or countercyclical? In this article, we provide an analysis toward answering these questions, and, in doing so, reconcile the aforementioned two seemingly inconsistent views on CRAs. In a two-period reputation model, we show that (i) both rating inflation and rating deflation can occur in equilibrium, and the agency's behavior can be distorted due to either excessive or insufficient reputation concerns; and (ii) credit ratings are procyclical: rating inflation is more likely to occur in a boom, while rating deflation is more likely to occur in a recession.

In our model, a CRA can be either an honest or an opportunistic type; an investor can be either sophisticated or naive; and a security can be either good or bad. The CRA receives a noisy signal about the quality of the security and issues a good or a bad rating upon the request of the issuer. The issuer will pay for and publish the good rating, but not the bad one. The honest CRA always reports the true signal, while the opportunistic CRA chooses the rating to maximize its expected payoffs. The sophisticated investors update their beliefs rationally, while the naive investors take the ratings at face value.

The opportunistic CRA faces the trade-off between the current benefit, which is the rating fees paid by the issuer upon receiving a favorable rating, and the future reputation costs. If the reputation costs are sufficiently small, the CRA will inflate the rating; if the reputation costs are sufficiently large, the CRA will deflate the rating in order to preserve the reputation; only when the reputation costs are in the intermediate range, the CRA will rate truthfully. We then relate the result to business cycles. During the boom, the default probability of the security is low, thus the reputation loss of lying is low, and the opportunistic CRA tends to inflate the rating. During the recession, the default probability of the security is high, thus the reputation costs are high when a good-rating security fails, and the CRA more likely deflates the rating to preserve its reputation.

Our paper complements the existing literature in two ways. First, besides rating inflation, we characterize the rating-deflation result. Scrutiny on CRAs during the recent financial crisis has generated many new studies. Most of them focus on the issue of rating inflation. Bolton et al. (2012) combine three sources for rating inflation: CRAs understating risk to attract business, issuers' rating shopping behavior, and the existence of trusting investors. Mathis et al. (2009) argue that reputation concerns are not sufficient to discipline rating agencies. When rating structured finance product becomes the major profit source, the CRA always inflates the rating with a positive probability. Skreta and Veldkamp (2009) attribute the reasons for rating inflation to rating shopping and security complexity. Rating shopping allows the issuer to disclose only the most favorable one, and increases in the complexity of the financial assets can create a systematic bias in disclosed ratings, despite that the CRAs produce unbiased estimate ratings.[2] Complementary to this literature, we find that in addition to rating inflation, rating deflation is also possible in equilibrium; excessive reputation concern can also distort the agency's behavior, just like insufficient reputation concern, albeit toward the opposite direction.

Second, we show that rating inflation is more likely to happen in a boom, while rating deflation is more likely to happen in a recession. In other words, the credit rating is procyclical. Bolton et al. (2012) also show that rating inflation is more likely during booms. This happens in their model because investors are more optimistic, while our reason is that the default probability of the security is lower during booms than during recessions. Bar-Isaac and Shapiro (2013) and Fulghieri et al. (2014) also discuss credit ratings over business cycles. Both of them demonstrate that rating quality is countercyclical: a CRA is more likely to issue more accurate ratings in a recession than in a boom. In other words, the rating quality can be better in a recession than in a boom. We show, however, that the rating quality can be biased in both states: it is upward-biased in a boom and downward-biased in a recession.

Our procyclical rating result is consistent with empirical works. Several papers have documented evidences that ratings inflation is more likely to happen during booms. Ashcraft et al. (2010) point out the issuance volume of MBS went up sharply between 2005–2007 while rating accuracy decreased, and later rating downgrades for the 2005–2007 cohorts were significantly larger than for the previous one. Auh (2013) finds that rating standards are procyclical. Firms receive overly pessimistic ratings in a recession, compared to during a boom. Benmelech and Dlugosz (2009) show that there were massive pre-crisis upgrading compared to the massive downgrading during the subprime crisis. Ferri et al. (1999) demonstrate that during the East Asian financial crisis, CRAs' ratings were procyclical. Having failed to predict the emergence of the crisis, CRAs became excessively conservative. They downgraded East Asian crisis countries more than these countries' economic fundamentals would justify.

The rest of the paper is organized as follows. Section 2 presents a model of a monopolistic CRA. Section 3 shows how both rating inflation and rating deflation may occur in equilibrium. Section 4 introduces economic states to the model and demonstrates that ratings are procyclical. Section 5 concludes.

2. Model

Consider a model that builds on Bolton et al. (2012). There are three kinds of risk-neutral agents: a CRA, issuers, and a unit mass of investors. Issuers seek external funding by selling securities to investors. There are two types of securities: good

[2] There are other related theoretical works. Mariano (2012) shows that ratings might not reflect private information or even contradict public information when CRAs attempt to minimize reputation costs. Opp et al. (2013) predict variation in rating quality across asset classes. They address the reason why rating standard were getting more lenient for structured products, whereas the rating of corporate bonds seemed to stay conservative. See also Bar-Isaac and Shapiro (2011), Cohn et al., 2013, Frenkel (2015), Goel and Thakor (2011), and Manso (2013).

($\tilde{g}$) or bad ($\tilde{b}$). Good securities do not fail, while bad ones fail with probability p. Both types of securities generate the same return R if not fail, and zero otherwise. All investors and issuers believe ex ante that a security is good with probability $\frac{1}{2}$. Assume $(1-\frac{p}{2})R<r$, where r is the reservation return that the investors need for one unit of investment. Thus, without further information, the investors are not willing to buy the security.

There are two periods, and there is an issuer in each period. At the beginning of period 1, in order to sell the security, the issuer approaches the CRA for a rating. The CRA first posts a fee ϕ, then receives a private signal $\tau \in \{g,b\}$ with the following information content:

$$\Pr(g \mid \tilde{g}) = \Pr(b \mid \tilde{b}) = \mu, \quad \text{where} \quad \frac{1}{2} < \mu < 1.$$

After that, the CRA produces a credit rating: $m=G$ (good) or $m=B$ (bad). Assume $(1-(1-\mu)p)R>r$; that is, an investor with reliable information that the security is good is willing to purchase it. After observing the rating, the issuer chooses either to buy and publish the rating or not to do so.

Proportion β of the investors are naive, while the rest are sophisticated, with $0<\beta\leq 1$. Both types of investors observe the rating published. The sophisticated investors know that the CRA can be either the honest type (H) or the opportunistic type (O), and the ex ante probability of having an honest type is η. But the naive investors regard the CRA trustworthy and thus take its rating at face value. The honest CRA always reports the true signal it receives, whereas the opportunistic CRA reports the rating which maximizes its expected payoffs.

If the security is issued, at the end of the period, both types of investors observe whether it is a success or a failure. The sophisticated investors will then find out whether or not the CRA lied by checking the report, the data and the facts.[3] If the CRA is found lying, the sophisticated investors know that the CRA is an opportunistic type for sure. If the CRA is not found lying, the sophisticated investors update their beliefs about the CRA's type accordingly.[4] But the naive investors remain naive: Once the "G" rating security failed, they punish the CRA by ignoring its future reports. Thus, the naive investors are naive both ex ante and ex post: ex ante they take the rating at face value, and ex post they take the result of the security at face value as well.[5] One way to rationalize the naive investors' behavior is by the repeated game setting, where the investors use the grim-trigger strategy: they will trust the CRA as long as the "G" rating securities succeed; once the "G" rating assets fail, they will not trust the CRA any more.[6]

An investor can purchase either one or zero unit of the security. Define

$$v \equiv (1-(1-\mu)p)R - r. \tag{1}$$

v is the value of a security to a naive investor when the CRA reports $m=G$ to that security, whether truthfully or not. Also, let

$$\omega_t \equiv R(1-(1-\alpha_t)p) - r, \tag{2}$$

where $\alpha_t \equiv \Pr(\tilde{g} \mid G, t)$ is a sophisticated investor's posterior belief that the project is good after observing a "G" rating at period t, $t=1,2$. ω_t is the expected value of a security to the sophisticated investor when the CRA reports $m=G$.

Following Laffont and Tirole (1993) and Bolton et al. (2012), the second-period payoff is weighted by a parameter δ, which can be larger than 1. As in Bolton et al. (2012), the parameter value δ represents the importance of the future reputation relative to the current gains for the CRA.

The timing of the game is as follows:

1. At the beginning of period 1, the issuer approaches the CRA for a rating for its security.
2. The CRA posts its rating fee ϕ, then receives a private signal, and makes a rating of $m=G$ or $m=B$.
3. The issuer receives the rating and decides whether to buy and publish it. If the security is not issued, there is zero payoff to all parties in period 1.
4. If the security is issued, investors observe its price and rating, and each decides whether to purchase one unit or not. At the end of period 1, the investment outcome, success or failure, is realized.
5. The game then moves to the beginning of period 2, whether or not a security is issued in period 1. In the second period, the game in the first period is repeated.

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3 If the security is not issued at all, then it is impossible for the investors to find out whether the CRA lied or not.

4 Bolton et al. (2012) assume that investors, both the sophisticated and the naive, can find out ex post whether the CRA lied in the event of a default. Here, we assume that the sophisticated investors can find out whether the CRA lied ex post. This assumption simplifies the analysis, but we fully acknowledge that it is somewhat extreme (see also Bolton et al., 2012 footnote 15 for a justification).

5 There are quite a few empirical studies on naive investors. For example, Malmendiera and Shanthikumar (2007) show that small investors follow security analysts' recommendations and reports literally.

6 See Bolton et al. (2012) and others (for example, Hirshleifer, 2001; Barberis and Thaler, 2003) for more discussion and justification about the behavior of the naive or irrational investors.

Table 1
Decision making process of opportunistic CRA.[7]

	Nature	Precision	Private signal	Probability	Rating
Good	$\tilde{g}$	$\xrightarrow{\mu}$	g	$\xrightarrow{y}$	G
Bad	$\tilde{b}$	$\xrightarrow{\mu}$	b	$\xrightarrow{z}$	B

Let y (resp. z) be the probability that the opportunistic CRA gives a G (resp. B) rating to a security with good (resp. bad) signal in the first period, then $y, z \in [0, 1]$. We can show the decision making process of opportunistic CRA in Table 1.

3. Equilibrium analysis

3.1. Preliminaries

To facilitate the equilibrium characterization, we start by considering how the investors will update their beliefs and how the CRA will charge fees with different ratings in different periods. As it will become clear later, the opportunistic CRA will always give G rating in period 2.

3.1.1. Period 1

Note first that the issuer will not pay for a B rating, which makes the rating fee for a B rating equal to zero. Thus, when observing no rating (N) for the first period, the sophisticated investors infer that the rating must be a B, and they update their beliefs about the CRA's type using this information:[8]

$$\eta_1^N \equiv \Pr(H \mid N) = \Pr(H \mid B) = \frac{\eta}{\eta + (1-\eta)(z+1-y)}.$$

Next, when observing a G rating in period 1, the sophisticated investors believe that the security is a good type with probability:

$$\alpha_1 \equiv \Pr(\tilde{g} \mid G) = \frac{\mu\eta + [\mu y + (1-\mu)(1-z)](1-\eta)}{\eta + (1-\eta)(y+1-z)},$$

and the sophisticated investors are willing to pay a price no more than $\omega_1 = R(1-(1-\alpha_1)p) - r$ to purchase it.

Notice that the issuer can post only a single price. When receiving a G rating, the issuer will post either price v to sell to the naive investors, or price $\omega_1 \le v$ to sell to all investors, with payoffs βv and ω_1, respectively. Under our assumption that the monopoly CRA can extract all the surplus from the issuers, the rating fee that CRA can charge for a G rating in the first period is:

$$\phi_1 = \max\{\beta v, \omega_1\}.$$

At the end of the first period, for a G-rated security (which is issued), the investment outcome, is realized and observed by all players. The naive investors will continue to believe the CRA if they see a success, and otherwise they will no longer pay attention to the CRA's ratings. For the sophisticated investors, if the CRA lied (i.e., reported G when the signal was b), their updated belief on the probability that the CRA is an honest type becomes $\eta_1^L = 0$; if the CRA did not lie, their updated belief, when there is a success (S) or a failure (F), is:

$$\eta_1^S = \eta_1^F \equiv \Pr(H \mid G, g) = \frac{\eta}{\eta + y(1-\eta)}.$$

3.1.2. Period 2

In the second period, there is no more reputation concern so that the opportunistic CRA always inflates the rating. When observing a G rating, the sophisticated investors update beliefs that the security is a good type:

$$\alpha_2^i \equiv \Pr(\tilde{g} \mid G) = \frac{\mu\eta_1^i + (1-\eta_1^i)}{2-\eta_1^i}, i = N, L, S, F,$$

[7] We thank an anonymous referee for offering this vivid table.
[8] Please see Appendix A for the calculation details of all of the following updating beliefs.

Table 2
Summary of notation.

$\tilde{g}, \tilde{b}$	Good and bad types of security, respectively.
g, b	Good and bad private signals, respectively.
G, B	Good and bad ratings, respectively.
H, O	Honest and opportunistic types of CRA, respectively.
R	Return of securities if not failed.
r	Reservation return of one unit of investment for investors.
p	Default probability of bad securities.
μ	Precision of received private signal.
β	Proportion of naive investors.
η	Sophisticated investor's belief of having an honest-type CRA
y, z	Probability of opportunistic CRA truthfully rate G and B, respectively.
α_t	Sophisticated investor's posterior belief on a G-rated security to be a good type at period t.
δ	Reputation parameter.
v	Value of G-rated security to naive investors.
ω_t	Value of G-rated security to sophisticated investors at period t.
ϕ_t	Rating fee at period t.
w	Value of a G-rated security to sophisticated investors in the second period if the CRA was truthful in the first period.
k	Value of a G-rated security to sophisticated investors in the second period if the CRA gave a B-rated in the first period.

corresponding to the four possible situations in period 1: no security was issued (N); the CRA was caught lying (L); the CRA reported truthfully and the security was a success (S) or a failure (F). The probability that the sophisticated investors assign to the security in period 2 as being a good type is:

$$\alpha_2^N = \frac{\eta\mu + (1-\eta)(z+1-y)}{\eta + 2(1-\eta)(z+1-y)}, \quad \alpha_2^L = \frac{1}{2}, \quad \alpha_2^S = \alpha_2^F = \frac{\eta\mu + (1-\eta)y}{\eta + 2(1-\eta)y}.$$

Therefore, the fees charged in period 2 by the CRA are, with $\omega_2^i = R(1-(1-\alpha_2^i)p) - r$ for $i = S, F, L, N$:

- $\phi_2^S = \max\{\beta v, \omega_2^S\}$ if the first period outcome was a success and the CRA did not lie;
- $\phi_2^{LS} = \beta v$ if the first period outcome was a success and the CRA lied;
- $\phi_2^F = (1-\beta)\omega_2^F$ if the first period outcome was a failure and the CRA did not lie; and
- $\phi_2^N = \max\{\beta v, \omega_2^N\}$ if in the first period the investors observed no rating.

For notation purpose, we define

$$w \equiv R\left\{1 - \left[1 - \frac{1-(1-\mu)\eta}{2-\eta}\right]p\right\} - r, \tag{3}$$

$$k \equiv R\left\{1 - \left[1 - \frac{\eta\mu + 2(1-\eta)}{4-3\eta}\right]p\right\} - r. \tag{4}$$

w is the expected value of a G-rated security to the sophisticated investor in the second period, given the CRA was truthful in the first period; and k is the expected value of a sophisticated investor for a G -rated security in the second period, when the CRA gave a B-rated in the first period.[9] Notice that $\frac{1}{2} < \frac{\eta\mu+2(1-\eta)}{4-3\eta} \leq \frac{1-(1-\mu)\eta}{2-\eta} < \mu$. We assume $k > 0$ for the rest of the analysis. It follows that $w \geq k > 0$.

We summarize the notation in Table 2.

3.2. Equilibrium

We will first discuss three pure-strategy equilibria: (1) truthful rating ($y = 1, z = 1$), where the opportunistic CRA always reports the true signal in period 1; (2) rating inflation ($y = 1, z = 0$), where the opportunistic CRA gives G rating in period 1 regardless of the signal; and (3) rating deflation ($y = 0, z = 1$), where the opportunistic CRA issues rating $m = B$ in period 1 no matter what signal it receives.[10] Later on, we will also analyze two mixed-strategy equilibria: semi-inflation ($y = 1, z \in (0, 1)$) and semi-deflation ($y \in (0, 1), z = 1$), which will be defined later.

[9] The economic meanings of w and k will become clearer in later analysis.
[10] The case of false reporting ($y = 0, z = 0$) is impossible for the CRA.

3.2.1. Truthful rating, rating inflation and deflation

Suppose that the CRA reports truthfully in equilibrium, that is, $y = 1, z = 1$. The sophisticated investors' beliefs are consistent with the CRA's strategy. Therefore, in the first period, the sophisticated investors believe that a *G*-rated security is a good type with probability μ, and they are willing to pay a price no more than v to buy it. Then the CRA will charge rating fee v for a "*G*" rating. At the end of the first period, the sophisticated investors' belief remains η. In the second period, the sophisticated investors' belief for a *G*-rated security is $\frac{1-(1-\mu)\eta}{2-\eta}$ and their willingness to pay for it is no more than w, where w is defined in Eq.(3). The CRA will charge fee $\phi_2^{S,T} = \phi_2^{N,T} = \max\{\beta v, w\}$ or $\phi_2^{F,T} = (1-\beta)w$, depending on the first period outcome *S*, *N*,or *F*.

With the truthful-rating strategy, conditional on receiving a good signal of the security, the CRA will report "*G*" and earn:

$$\pi^T(G \mid g) = \phi_1^T + \delta\left\{[1-(1-\mu)p]\phi_2^{S,T} + (1-\mu)p\phi_2^{F,T}\right\} = v + \delta\{\max\{\beta v, w\} + (1-\mu)p[(1-\beta)w - \max\{\beta v, w\}]\}$$

as the security will succeed with probability $1-(1-\mu)p$ and fail with probability $(1-\mu)p$. Conditional on receiving a bad signal, the CRA will report *B*, and its payoff will be:

$$\pi^T(B \mid b) = \delta\phi_2^{N,T} = \delta\max\{\beta v, w\}.$$

If the CRA deviates to report "*B*" when it receives a "*g*" signal, its payoff is:

$$\pi^T(B \mid g) = \delta\phi_2^{N,T} = \delta\max\{\beta v, w\}.$$

If it deviates to report "*G*" when it receives a "*b*" signal, its payoff is:

$$\pi^T(G \mid b) = \phi_1^T + \delta(1-\mu p)\beta v = v + \delta(1-\mu p)\beta v.$$

By comparing the above four payoffs to make sure that the CRA will report truthfully, we have thus established:

Lemma 1. *There exist values* $\underline{\delta}^T \equiv \frac{v}{\max\{\beta v, w\} - (1-\mu p)\beta v}$ *and* $\bar{\delta}^T \equiv \frac{v}{(1-\mu)p[\max\{\beta v, w\} - (1-\beta)w]}$, *with* $\underline{\delta}^T < \bar{\delta}^T$, *such that truthful rating is the equilibrium strategy for the opportunistic CRA if and only if* $\underline{\delta}^T \le \delta \le \bar{\delta}^T$.

Proof. Please see Appendix A.■

Lemma 1 says that when the reputation parameter δ is in the intermediate range ($\underline{\delta}^T \le \delta \le \bar{\delta}^T$), the opportunistic CRA will rate truthfully.

By the same token, we have the following two Lemmas for rating inflation and rating deflation, respectively:

Lemma 2. *Rating inflation is the equilibrium strategy for the opportunistic CRA if and only if* $\delta \le \delta^I$, *where* $\delta^I \equiv \frac{\max\{\beta v, w\}}{v-(1-\mu p)\beta v}$ *and* $0 < \delta^I \le \underline{\delta}^T$.

Proof. Please see Appendix A.■

Lemma 2 means when the reputation parameter δ is sufficiently small ($\delta \le \delta^I$), the opportunistic CRA will inflate ratings. Given δ is very small, the CRA has little reputation concern and it does not care too much about future payoff. Instead, the CRA has incentive to inflate ratings for current gains.

Lemma 3. *Rating deflation is the equilibrium strategy for the opportunistic CRA if and only if* $\delta \ge \delta^D$ *given* $\beta > \hat{\beta}$, *where* $\hat{\beta} \equiv \frac{1}{1+(1-\mu)p}$ *and* $\delta^D \equiv \frac{v}{\max\{\beta v, k\} - v + (1-\mu)p\beta v} \ge \bar{\delta}^T$.

Proof. Please see the Appendix A.■

Lemma 3 states that given both the reputation factor δ and the proportion of naive investors β are sufficiently large ($\delta \ge \delta^D$ and $\beta > \hat{\beta}$), the opportunistic CRA will deflate ratings. As δ is very large, the CRA cares about the future payoffs. Given the CRA's private signal is imperfect and the proportion of naive investors is sufficiently high, the CRA always wants to give a "*B*" rating in the first period in order to guarantee its second-period payoffs from the naive investors.

Generally speaking, reputation concern can help to discipline the agent's behavior. However, Lemma 3 shows that excessive reputation concern can distort the agent's behavior and lead to the inefficient outcome. In our case, too much stake in the future makes the CRA behave very conservatively and issue downward-biased ratings in order to preserve reputation.

3.2.2. Equilibrium analysis

Summarizing the findings from Lemmas 1, 2 and 3, we have our first main result below, providing the equilibrium characterization:

Proposition 1. *There exist five critical values* δ^I,$\underline{\delta}^T$,$\bar{\delta}^T$, δ^D, *and* $\hat{\beta}$, *with* $\delta^I \leq \underline{\delta}^T < \bar{\delta}^T \leq \delta^D$, *such that in equilibrium the opportunistic CRA has the following first-period strategies: (i) it inflates the rating if and only if* $\delta \in [0, \delta^I]$, *setting rating fee* $\phi_1^I = \max\{\beta v, w\}$; *(ii) it reports the true signal if and only if* $\delta \in \left[\underline{\delta}^T, \bar{\delta}^T\right]$, *setting rating fee* $\phi_1^T = v$; *and (iii) provided that* $\beta > \hat{\beta}$, *it deflates the rating if and only if* $\delta \in [\delta^D, +\infty)$, *setting rating fee* $\phi_1^D = 0$.

When choosing rating strategy, the opportunistic CRA faces a trade-off between the current benefit and the future reputation cost. Proposition 1 shows that when the reputation parameter δ is sufficiently low such that the reputation loss will be small, the CRA inflates the rating; when the reputation parameter and the proportion of naive investors are sufficiently high such that the reputation loss is large, the CRA deflates the rating in order to preserve the reputation; only when the reputation parameter is in the intermediate range, the CRA provides truthful ratings.

The following results show how changes in key parameters of our model, β, μ, and p may affect the equilibrium outcome.

Corollary 1. *(i) The opportunistic CRA is less likely to rate truthfully when the proportion of naive investors* (β) *is higher.*[11] *(ii) Suppose that* $\beta > \frac{w}{v}$. *Then the opportunistic CRA is more likely to provide truthful rating when the private signal is more accurate (i.e.,* μ *is higher).*

Proof. Please see Appendix A.■

In the rating-inflation regime, the parameter value β determines the potential revenue from the naive investors. Keeping other things constant, a higher β implies a larger $\delta^I \left(\frac{\partial \delta^I}{\partial \beta} > 0\right)$, and hence there is a larger region of parameter values δ under which the CRA inflates the rating. In the rating-deflation regime, the parameter value β determines the punishment intensity if a G-rated security fails. Holding other things constant, a higher β implies a smaller $\delta^D \left(\frac{\partial \delta^D}{\partial \beta} < 0\right)$, and thus there is a larger region of parameter values under which the CRA deflates the rating. Therefore, the opportunistic CRA is less likely to rating truthfully when the portion of naive investors (β) is higher.

When the private signal is more accurate (μ is higher), the CRA will be more likely to report the true signal, provided that the proportion of naive investors is sufficiently large ($\beta > \frac{w}{v}$). The reason is because a more accurate private signal implies a higher cost of mis-reporting, increasing the incentive for truthful rating.

Corollary 2. *Suppose that* $\beta > \frac{w}{v}$. *The opportunistic CRA is more likely to inflate the rating when the default probability* p *is small; more likely to deflate the rating when the default probability is large.*

Proof. Please see Appendix A.■

Corollary 2 says that given the proportion of naive investors is sufficiently large, a larger default probability (p) leads opportunistic CRA more likely to deflate the rating. Due to the noisy signal, when the default probability is large, both rating inflation and rating truthfully have a large probability to be punished by the naive investors. Therefore, the CRA is more likely to deflate the rating. On the other hand, when the default probability is small, rating inflation is more likely since the chance of being caught lying and being punished is also small.

Notice that the default probability of projects p is usually much lower during booms than during recessions. Therefore, it might be the case that rating inflation is more likely to happen in booms when p is small, while rating deflation is more likely to happen during recessions when p is large. We will formalize this observation in Section 4.

3.2.3. Semi-inflation and semi-deflation

Here, we consider two mixed-strategy equilibria: (1) semi-inflation ($y = 1, z \in (0,1)$), where the opportunistic CRA gives G rating if it receives good signal, and gives B rating with probability z if it receives bad signal; and (2) semi-deflation ($y \in (0,1), z = 1$), where the opportunistic CRA issues rating B if it receives bad signal, and issues G rating with probability y if it receives good signal.

Lemma 4. *For any* $\delta \in (\delta^I, \underline{\delta}^T)$, *there exists a unique* $z^*(\delta) \in (0,1)$, *such that, semi-inflation (i.e.,* $(y,z) = (1, z^*(\delta))$*) is the equilibrium strategy for the opportunistic CRA.*

[11] *The rating deflation argument is within interval* $\beta \in (\widehat{\beta}, 1]$.

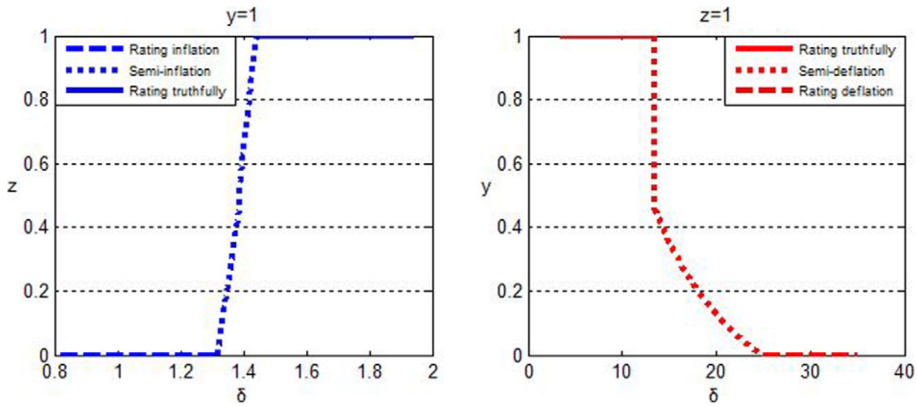

Fig. 1. Numerical outcomes.

Proof. Please see Appendix A.■

Lemma 4 means in between the two pure strategy equilibrium: the rating inflation and the truthful rating, there exists a mixed strategy equilibrium, the semi-inflation equilibrium. Notice that $z^*(\delta)$ is an increasing function of δ, that is, as reputation cost δ increases, the CRA is less likely to inflate the rating.

Lemma 5. *For any $\delta \in (\bar{\delta}^T, \delta^D)$, there exists a unique $y^*(\delta)$, such that, semi-deflation (i.e., $(y,z) = (y^*(\delta), 1)$) is the equilibrium strategy for the opportunistic CRA.*

Proof. Please see Appendix A.■

Lemma 5 indicates for any $\delta \in (\bar{\delta}^T, \delta^D)$, there exists a unique belief $y^*(\delta)$ supporting the CRA's semi-deflation strategy. Note that $y^*(\delta)$ is a decreasing function of δ. It means as reputation costs increase, the CRA will be more likely to deflate the rating.

3.2.4. A numerical example

The following is one numerical example to illustrate the results of both the pure-strategy equilibria and the mixed-strategy equilibria. Let $\mu = 0.9, \eta = 0.96, p = 0.8, \beta = 0.9630, R = 1, r = 0.6$. Hence, we have $\delta^I = 1.3185, \underline{\delta}^T = 1.4423, \bar{\delta}^T = 13.4586$, $\delta^D = 25$.

In Fig. 1, the horizontal axis is the reputation parameter δ. The vertical axis of the left-side graph is $z \in [0,1]$: the CRA's probability to give B rating if it receives bad signal, given $y = 1$. And the vertical axis of the right-side graph indicates $y \in [0,1]$: the CRA's probability to give G rating if it receives good signal, given $z = 1$.

From Fig. 1 we can see that:

1. When $\delta \in [0, 1.3185]$, $(y,z) = (1,0)$, or rating inflation is the equilibrium strategy.
2. When $\delta \in (1.3185, 1.4423)$, $(y,z) = (1, z^*(\delta))$, or semi-inflation is the equilibrium strategy.
3. When $\delta \in [1.4423, 13.4586]$, $(y,z) = (1,1)$, or truthful rating is the equilibrium strategy.
4. When $\delta \in (13.4586, 25.0000)$, $(y,z) = (y^*(\delta), 1)$, or semi-deflation is the equilibrium strategy.
5. When $\delta \in [25.0000, +\infty)$, $(y,z) = (0,1)$, or rating deflation is the equilibrium strategy.

As we can see from the graph, δ is the key parameter fixing the values of others. The CRA's equilibrium rating strategy is changing as δ shifts. When δ is small enough, the CRA does not care about reputation and inflates rating for sure. As δ increases, then the CRA will inflate rating with probability $z^* \in (0,1)$. Only when δ shifts to the certain intermediate range, the CRA will rate truthfully. However, if δ continues to increase, the CRA will deflate the rating with probability $y^* \in (0,1)$. If δ is big enough, the CRA will deflate the rating for sure. The key message is that reputation concern might *not* discipline the agent. Both excessive and insufficient reputation can distort the agency's behavior, although the distortions are toward different directions.

4. Ratings and business cycle

In this section, we introduce state variables to discuss the relationship between credit ratings and the business cycle. Suppose there are two states $s \in \{h, l\}$, where h corresponds to high economic activities or booms, and l to low economic activities or

Table 3
Summary of revised notation.

h	Economic booms.
l	Economic recessions.
s	State parameter, $s \in \{h, l\}$.
θ_s	Probability of current state s will remain in the next period.
p_s	Default probability of bad security in state s.
$\bar{v}_s$	Expected value of a naive investor for the G-rated security in the next period if the current state is s.
$\bar{w}_s$	Expected value of a sophisticated investor for the G -rated security in period 2 when the CRA was truthful in the first period and the current state is s.
$\bar{k}_s$	Expected value of a sophisticated investor for the G -rated security in period 2 when the CRA gave B-rated in the first period and the current state is s.

recessions. We assume that the probability of failure for the securities is lower during a boom than during a recession: $p_h < p_l$. It follows that $v_h > v_l$, $w_h > w_l$, and $k_h > k_l$. Everything else is assumed to be the same in the two states.

Let θ_s be the probability that the current state s will remain in the next period. Then $1 - \theta_s$ is the transition probability from the current state s to the other state. For $-s \in \{h, l\}$ and $-s \neq s$, if $\theta_s = 1 - \theta_{-s}$, the state in each period is an i.i.d draw from the same distribution; if $\theta_s > 1 - \theta_{-s}$, there is a positive correlation between states; and if $\theta_s < 1 - \theta_{-s}$, there is a negative correlation between states. A higher θ_s means a longer duration for the state s and a slow move to the other state. The following is the transition matrix:

$$\begin{array}{ccc} & h & l \\ h & \theta_h & 1-\theta_h \\ l & 1-\theta_l & \theta_l \end{array}$$

The transition matrix and the nature of the state in each period are assumed to be public information. Denote:

$$\bar{v}_s \equiv \theta_s v_s + (1-\theta_s)v_{-s},$$

$$\bar{w}_s \equiv \theta_s w_s + (1-\theta_s)w_{-s},$$

$$\bar{k}_s \equiv \theta_s k_s + (1-\theta_s)k_{-s}.$$

Then, given the current state s, $\bar{v}_s$ is the expected value of a naive investor for the G-rated security in the next period; $\bar{w}_s$ is the expected value of a sophisticated investor, when the CRA was truthful in the current period, for the G-rated security in the next period; and $\bar{k}_s$ is the expected value of a sophisticated investor, when the CRA gave a B-rating in the current period, for the G-rated security in the next period. Table 3 summarizes the revised notation.

We further define the following thresholds, for $s \in \{h, l\}$:

$$\delta_s^I \equiv \frac{\max\{\beta v_s, w_s\}}{\bar{v}_s - (1-\mu p_s)\beta\bar{v}_s},$$

$$\underline{\delta}_s^T \equiv \frac{v_s}{\max\{\beta\bar{v}_s, \bar{w}_s\} - (1-\mu p_s)\beta\bar{v}_s},$$

$$\bar{\delta}_s^T \equiv \frac{v_s}{(1-\mu)p_s\left[\max\{\beta\bar{v}_s, \bar{w}_s\} - (1-\beta)\bar{w}_s\right]},$$

$$\delta_s^D \equiv \frac{v_s}{\max\{\beta\bar{v}_s, \bar{k}_s\} - \bar{v}_s + (1-\mu)p_s\beta\bar{v}_s}.$$

Then, similarly as in the previous section, we have

$$\delta_s^I \le \underline{\delta}_s^T < \bar{\delta}_s^T \le \delta_s^D.$$

Also, let

$$\hat{\beta}_s \equiv \frac{1}{1+(1-\mu)p_s}, \text{ for } s = h, l,$$

then $\hat{\beta}_l < \hat{\beta}_h$. Using the same arguments leading to Proposition 1, we immediately have:

Lemma 6. *Suppose that the state in period 1 is $s \in \{h, l\}$. Then the opportunistic CRA's equilibrium strategy in period 1 is: (i) it inflates the rating if and only if $\delta_s \le \delta_s^I$, setting the rating fee $\phi_{1,s}^I = \max\{\beta v_s, w_s\}$; (ii) it reports the true signal if and only if $\underline{\delta}_s^T \le \delta_s \le \bar{\delta}_s^T$, setting $\phi_{1,s}^T = v_s$; and (iii) provided that $\beta > \hat{\beta}_h$, it deflates the rating if and only if $\delta_s \ge \delta_s^D$, setting $\phi_{1,s}^D = 0$.*

Similar to Proposition 1, when the reputation parameter δ_s is sufficiently small, the CRA inflates the rating; when the reputation parameter and the proportion of naive investors are sufficiently large ($\beta > \hat{\beta}_h$), the CRA deflates the rating; when the reputation parameter is in the intermediate range, the CRA reports the true signal.

Proposition 2 below states our second main result, connecting ratings to business cycles.

Proposition 2. *Rating inflation is more likely to happen in a boom; and there exists $\bar{\beta} \in (0,1)$, such that for $\beta > \bar{\beta}$, rating deflation is more likely to happen in a recession.*

Proof. Please see Appendix A.■

Regardless of whether the states are independent or correlated across periods, we have $\delta_h^I > \delta_l^I$ and $\delta_h^D > \delta_l^D$; that is, the range of parameter values for the equilibrium of rating inflation is larger in a boom, whereas the range of parameter values for the equilibrium of rating deflation is larger in a recession. In this sense, CRA's credit ratings are procyclical: rating inflation is more likely to happen in a boom, while rating deflation is more likely to happen in a recession.

The intuition for this procyclical rating result is as follows. Since the probability of failure for the bad security is lower in a boom than in a recession ($p_h < p_l$), the expected payoff of issuing the security (and hence also the CRA's rating fee) is higher in the boom than in the recession ($v_h > v_l$, $w_h > w_l$, and $k_h > k_l$). Consequently, relative to in a recession, the current gain from rating inflation is higher in a boom, while the expected reputation cost of rating inflation is lower as the default probability is lower in a boom. Therefore, the opportunistic CRA has more incentive to inflate the ratings in a boom. On the other hand, the current loss from rating deflation is lower in a recession than in a boom, keeping other things constant. Thus, the opportunistic CRA is more likely to deflate ratings in a recession in order to reap the future gain.

Corollary 3. *(i) Rating inflation is more likely to happen in the first period when the second period is more likely to be a recession; (ii) Rating deflation is more likely to happen in the first period when the second period is more likely to be a boom.*

Proof. Please see Appendix A.■

Keeping other things unchanged, a larger recession probability in the second period means less payoffs for the CRA in that period, thus the CRA is more likely to inflate the rating in the first period. On the other hand, a larger booming probability means more revenue in the second period, thus the CRA is more likely to deflate the rating in the first period.

5. Conclusion

CRAs have been under intense scrutiny since recent global financial crisis. Existing literature as well as business practitioners have argued either that CRAs engage in rating inflation, or that they deflate ratings. This article provides an analysis that reconciles the two opposing views. We find that both rating inflation and rating deflation can occur in equilibrium. Furthermore, we find that credit rating is procyclical: rating inflation is more likely to happen in a boom while rating deflation is more likely to happen in a recession.

As a response to the CRAs' moral hazard problem, the US government has attempted to improve or tighten the regulation. Subtitle C in Title IX of The Dodd–Frank Wall Street Reform and Consumer Protection Act focuses entirely on the regulation of CRAs, referred to as Nationally recognized Statistical Rating Organizations (NRSROs). Some key elements of the provisions in Title IX Subsection C are:

- Disclosure. NRSROs are required to disclose their rating track records, their rating methodologies, and their use of third parties for due diligence efforts.
- Liability. Investors can bring private rights of action against CRAs for a knowing or reckless failure to conduct a reasonable investigation of the facts or to obtain analysis from an independent source.
- Deregister. The SEC has the authority to deregister any agency for providing bad ratings over time.

These requirements are likely to increase CRAs' efforts and reduce the moral hazard problem. For example, the disclosure principle allows investors to have more information, and thus to make more rational judgements when using credit ratings. However, these requirements mainly target rating inflation, and may exacerbate the problem of rating deflation. Facing legal liability, CRAs may reduce the number of ratings as well as increase the downward bias in ratings (Goel and Thakor, 2011), which could hurt the issuers. It seems that more studies are needed with regard to the consequences of the legislation.

There are several limitations of our paper. Our rating-deflation result depends on the existence of naive investors. It would be interesting to rationalize the naive investors' behavior, possibly in a repeated game setting. Also, we have assumed that the B-rated security will not be purchased. It could be interesting to extend this to settings where there are several rating classes, and only the lowest rating is the one that is not purchased. The rating deflation result, if it can still exist, is likely to have more restrictions on the parameter range of δ.[12]

Our reputation model has only two periods. In the infinitely repeated games with imperfect monitoring, Cripps et al. (2004) show that the reputation effect is necessarily transient. However, there also exists a small but new literature on reputation dynamics to explain the long-run reputation, in which the reputation is accumulated, consumed, and restored (Liu, 2011). Built on this literature, one might show that there exists a reputation cycle accompanied by a rating cycle. In particular, when the reputation of the CRA is above certain threshold, its rating standard is loose and cozy to milk the reputation; but when its reputation is below some threshold, its rating standard will be tight and restrictive to build up or rescue its reputation.

Most of the recent literature, both theoretical and empirical, focus on the rating inflation occurred before the financial crisis. But we do observe the phenomenon that when the crisis started, the CRAs became much more conservative by massively downgrading the ratings. For example, in 2007, as housing prices began to tumble, Moody's downgraded 83 percent of the \$869 billion in mortgage securities it had rated at the AAA level in 2006; on August 5, 2011, S&P downgraded U.S. debt for the first time in U.S. history, by one notch from AAA to AA+; since the spring of 2010, one or more of the Big Three relegated Greece, Portugal, and Ireland to "junk" status, and in January 2012, amid continued Eurozone instability, S&P downgraded nine eurozone countries, stripping France and Austria of their triple-A ratings.

Is the downgrading merely a correction to the previous rating inflation? Or does it involve rating deflation, downgrading more than the fundamentals would justify? Are ratings procyclical? Are there reputation and rating cycles? Given that credit ratings serve as public coordinating devices and that downgrading has major impacts in financial markets (Boot et al., 2006; Manso, 2013), this article calls for more research on these critical issues.

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Appendix A.

A.1. Beliefs

-
$$\begin{aligned}\eta_1^N &\equiv \Pr(H \mid N) = \Pr(H \mid B)\\ &= \frac{\Pr(B \mid H)\Pr(H)}{\Pr(B \mid H)\Pr(H) + \Pr(B \mid O)\Pr(O)}\\ &= \frac{\eta}{\eta + (1-\eta)(z+1-y)},\end{aligned}$$

-
$$\begin{aligned}\alpha_1 \equiv \Pr(\tilde{g} \mid G) &= \frac{\Pr(G \mid \tilde{g})\Pr(\tilde{g})}{\Pr(G \mid \tilde{g})\Pr(\tilde{g}) + \Pr\left(G \mid \tilde{b}\right)\Pr\left(\tilde{b}\right)}\\ &= \frac{\mu\eta + [\mu y + (1-\mu)(1-z)](1-\eta)}{\eta + (1-\eta)(y+1-z)},\end{aligned}$$

-
$$\begin{aligned}\eta_1^S = \eta_1^F \equiv \Pr(H \mid G, g) &= \frac{\Pr(G, g \mid H)\Pr(H)}{\Pr(G, g \mid H)\Pr(H) + \Pr(G, g \mid O)\Pr(O)}\\ &= \frac{\eta}{\eta + y(1-\eta)}.\end{aligned}$$

[12] We extend our monopolistic CRA model to the duopoly case in Appendix A. Both rating inflation and rating deflation remain, and truthful rating becomes less likely (see also Bolton et al., 2012).

- $$\begin{aligned}\alpha_2^i \equiv \Pr(\tilde{g} \mid G) &= \frac{\Pr(G \mid \tilde{g})\Pr(\tilde{g})}{\Pr(G \mid \tilde{g})\Pr(\tilde{g}) + \Pr\left(G \mid \tilde{b}\right)\Pr\left(\tilde{b}\right)} \\ &= \frac{\mu\eta_1^i + (1-\eta_1^i)}{2-\eta_1^i}, i = N, L, S, F,\end{aligned}$$

A.2. Proofs

Proof. (Lemma 1). The CRA will report truthfully if and only if

$$\pi^T(G \mid g) - \pi^T(B \mid g) \geq 0, \tag{5}$$

and

$$\pi^T(B \mid b) - \pi^T(G \mid b) \geq 0. \tag{6}$$

Condition (5) holds if and only if

$$\delta \leq \frac{v}{(1-\mu)p[\max\{\beta v, w\} - (1-\beta)w]} \equiv \bar{\delta}^T, \tag{7}$$

and condition (6) holds if and only if

$$\delta \geq \frac{v}{\max\{\beta v, w\} - (1-\mu p)\beta v} \equiv \underline{\delta}^T. \tag{8}$$

Since $\mu > 1/2$ and

$$\max\{\beta v, w\} - (1-\beta)w < \beta v,$$

we have

$$\bar{\delta}^T > \frac{v}{(1-\mu)p\beta v} > \frac{v}{\mu p \beta v} \geq \underline{\delta}^T.$$

■

Proof. (Lemma 2). The CRA may choose to inflate ratings, that is, $y = 1, z = 0$. The sophisticated investors' beliefs are consistent with the CRA's strategy in equilibrium. Thus,

$$\begin{aligned}\alpha_1^I &= \frac{1-(1-\mu)\eta}{2-\eta}, \quad \omega_1^I = R[1-(1-\alpha_1^I)p] - r = w, \quad \phi_1^I = \max\{\beta v, w\}; \\ \eta_1^{S,I} &= \eta_1^{F,I} = \eta, \quad \alpha_2^{S,I} = \alpha_2^{F,I} = \frac{1-(1-\mu)\eta}{2-\eta}; \\ \omega_2^{S,I} &= \omega_2^{F,I} = R[1-(1-\alpha_2^{S,I})p] - r = w; \\ \phi_2^{S,I} &= \max\{\beta v, w\}, \quad \phi_2^{F,I} = (1-\beta)w,\end{aligned}$$

where w is defined in Eq. (3). With the rating-inflation strategy, conditional on a bad signal, the CRA will report G. In this case, the CRA can collect the rating fee, but it will be found lying by the sophisticated investors and thus they will not buy in period 2. Naive investors will only purchase in period 2 if the security succeeds in period 1 (which occurs with $1 - \mu p$). Therefore, the CRA's payoff will be:

$$\pi^I(G \mid b) = \phi_1^I + \delta(1-\mu p)\beta v = \max\{\beta v, w\} + \delta(1-\mu p)\beta v.$$

Also, with the rating-inflation strategy, the CRA will report $m = G$ conditional on a good signal, and the CRA's payoff will be:

$$\begin{aligned}\pi^I(G \mid g) &= \phi_1^I + \delta\left\{[1-(1-\mu)p]\phi_2^{S,I} + (1-\mu)p\phi_2^{F,I}\right\} \\ &= \max\{\beta v, w\} + \delta\left\{\max\{\beta v, w\} - (1-\mu)p[\max\{\beta v, w\} - (1-\beta)w]\right\}.\end{aligned}$$

In the rating-inflation regime, the sophisticated investors believe that the opportunistic CRA always reports G. Once they find that there is no reporting, they know it must be a B rating, and they hold the belief that the CRA giving the B-rating is an honest type. Therefore,

$$\alpha_2^{N,I} = \mu, \quad \omega_2^{N,I} = v, \quad \phi_2^{N,I} = v.$$

Then the payoff for the CRA to deviate to report B is:

$$\pi^I(B) = \delta\phi_2^{N,I} = \delta v.$$

Notice that $\pi^I(G \mid g) \geq \pi^I(G \mid b)$. Hence, rating inflation is an equilibrium strategy if and only if

$$\pi^I(G \mid b) - \pi^I(B) \geq 0,$$

which holds if and only if

$$\delta \leq \frac{\max\{\beta v, w\}}{v - (1 - \mu p)\beta v} \equiv \delta^I.$$

Notice that

$$0 < \delta^I < \underline{\delta}^T,$$

we thus have the Lemma.■

Proof. (Lemma 3). The CRA may choose to deflate ratings, that is, $y = 0, z = 1$. In such an equilibrium,

$$\phi_1^D = 0, \quad \alpha_2^{N,D} = \frac{\eta\mu + 2(1-\eta)}{4 - 3\eta},$$
$$\omega_2^{N,D} = R\left[1 - \left(1 - \alpha_2^{N,D}\right)p\right] - r = k, \quad \phi_2^{N,D} = \max\{\beta v, k\},$$

where k is defined in Eq. (4). Note that $\phi_1^D = 0$ since the issuer will not pay for a B rating. With the rating-deflation strategy, conditional on receiving a good signal, the CRA reports $m = B$, with payoff:

$$\pi^D(B \mid g) = \delta\phi_2^{N,D} = \delta\max\{\beta v, k\}.$$

Conditional on receiving a bad signal, the CRA also reports $m = B$, and its payoff is:

$$\pi^D(B \mid b) = \delta\phi_2^{N,D} = \delta\max\{\beta v, k\}.$$

In the rating-deflation regime, the CRA may deviate to report $m = G$. Since the sophisticated investors believe that the opportunistic CRA always deflates the ratings in this regime, once they see G report, they believe that the CRA is an honest type. Then

$$\alpha_2^{i,D} = \mu, \quad \omega_2^{i,D} = v, \quad \text{where} \quad i = S, F,$$
$$\phi_2^{S,D} = v, \quad \phi_2^{F,D} = (1-\beta)v.$$

Therefore, this deviation when receiving a g signal brings payoff:

$$\begin{aligned}\pi^D(G \mid g) &= v + \delta\left\{[1 - (1-\mu)p]\phi_2^{S,D} + (1-\mu)p\phi_2^{F,D}\right\} \\ &= v + \delta[v - (1-\mu)p\beta v].\end{aligned}$$

Or, after receiving a bad signal, the CRA may deviate to report $m = G$, and the payoff from this deviation is:

$$\pi^D(G \mid b) = v + \delta(1 - \mu p)\beta v.$$

Notice that $\pi^D(G \mid g) \geq \pi^D(G \mid b)$. Therefore, the rating-deflation strategy is an equilibrium strategy if and only if

$$\pi^D(B \mid g) - \pi^D(G \mid g) = \delta \max\{\beta v, k\} - \{v + \delta[v - (1-\mu)p\beta v]\} \geq 0,$$

or

$$\delta \max\{\beta v, k\} - v - \delta[v - (1-\mu)p\beta v] \geq 0. \tag{9}$$

Since $\max\{\beta v, k\} \geq \beta v$, $\max\{\beta v, k\} - v + (1-\mu)p\beta v > 0$ if

$$\beta v - v + (1-\mu)p\beta v > 0,$$

which holds if $\beta > \hat{\beta} \equiv \frac{1}{1+(1-\mu)p}$. Then, when $\beta > \hat{\beta}$, Eq.(9) holds if and only if

$$\delta \geq \frac{v}{\max\{\beta v, k\} - v + (1-\mu)p\beta v} \equiv \delta^D;$$

and δ^D exists for any given $\beta > \hat{\beta}$. It follows that

$$\delta^D \geq \bar{\delta}^T.$$

■

Proof. (Corollary 1). (i) It is easy to show that $\frac{\partial \delta^I}{\partial \beta} > 0$, $\frac{\partial \delta^D}{\partial \beta} < 0$, and $\frac{\partial (\bar{\delta}^T - \underline{\delta}^T)}{\partial \beta} < 0$. It follows that, when β is higher, the regions of parameter values that support the rating inflation equilibrium and the rating deflation equilibrium are larger, and the region that support truthful rating is smaller. (ii) When $\beta > \frac{w}{v}$, it is straightforward to verify that $\frac{\partial \delta^I}{\partial \mu} < 0, \frac{\partial \underline{\delta}^T}{\partial \mu} < 0$, $\frac{\partial \bar{\delta}^T}{\partial \mu} > 0$, and $\frac{\partial \delta^D}{\partial \mu} > 0$. Therefore the opportunistic CRA is more likely to provide truthful rating when μ is higher.■

Proof. (Corollary 2). When $\beta > \frac{w}{v}$, it is easy to verify that δ^I, $\left(\bar{\delta}^T - \underline{\delta}^T\right)$ and δ^D are all decreasing in p. Therefore, the opportunistic CRA is more likely to inflate the rating when the default probability (p) is small; is more likely to deflate rating when the default probability (p) is large.■

Proof. (Lemma 4). For any $\delta \in (\delta^I, \underline{\delta}^T)$, let z as the ex ante belief of sophisticated investors for opportunistic CRA giving B rating if it receives bad signal. Thus, the willingness to pay of sophisticated investors is $\omega_1(z)$ in period 1, and $\omega_2^{N,I}(z)$ in period 2. We have

$$\pi^I(G \mid b; z) = \max\{\beta v, \omega_1(z)\} + \delta(1-\mu p)\beta v;$$
$$\pi^I(B \mid b; z) = 0 + \delta \max\left\{\beta v, \omega_2^{N,I}(z)\right\}.$$

Note that $\omega_1(0) = w$, $\omega_2^{N.I}(0) = v$ (rating inflation); $\omega_1(1) = v$, $\omega_2^{N,I}(1) = w$ (truthful rating).

We prove the existence of $z^*(\delta)$ by using three arguments. First, the continuity, that is, both $\pi^I(G \mid b; z)$ and $\pi^I(B \mid b; z)$ are continuous on z. Second, the weakly monotonicity, that is, $\pi^I(G \mid b; z)$ is weakly increasing with z, and $\pi^I(B \mid b; z)$ is weakly decreasing with z. Third, the crossing property, that is, $\pi^I(G \mid b; 1) > \pi^I(B \mid b; 1)$ and $\pi^I(B \mid b; 0) > \pi^I(G \mid b; 0)$.

From the beliefs updating procedure above, we can easily conclude that belief α_1 and α_2^N are Bayesian and continuous on z, and thus $\pi^I(G \mid b; z)$ and $\pi^I(B \mid b; z)$ are continuous.

For period 1, we can show that $\frac{\partial \omega_1(z)}{\partial z} > 0$ and $\frac{\partial \pi^I(G|b;z)}{\partial z} \geq 0$. For period 2, we can show that $\frac{\partial \alpha_2^N}{\partial z} < 0$, $\frac{\partial \omega_2^N(z)}{\partial z} < 0$, and $\frac{\partial \pi^I(B|b;z)}{\partial z} \leq 0$. Therefore, given any δ, $\pi^I(G \mid b; z)$ is weakly increasing with z and $\pi^I(B \mid b; z)$ is weakly decreasing with z. In the other words, when receiving bad signal, the CRA has more incentive to inflate rating in period 1 if investors hold higher belief of truthful rating.

Furthermore, since $\delta \in (\delta^I, \underline{\delta}^T)$, we have

$$\pi^I(G \mid b; 1) = \pi^T(G \mid b) > \pi^T(B \mid b) = \pi^I(B \mid b; 1);$$
$$\pi^I(B \mid b; 0) = \pi^I(B \mid b) > \pi^I(G \mid b) = \pi^I(G \mid b; 0).$$

The former inequality comes from the equilibrium condition of truthful rating, and the latter comes from the equilibrium condition of rating inflation in the previous part.

Therefore, for any $\delta \in (\delta^I, \underline{\delta}^T)$, there exists $z^*(\delta)$ satisfies $\pi^I(G \mid b; z^*(\delta)) = \pi^I(B \mid b; z^*(\delta))$. Thus, semi-inflation $(y, z) = (1, z^*(\delta))$ is a PBE strategy.■

Proof. (Lemma 5). For any $\delta \in (\bar{\delta}^T, \delta^D)$, let y as the ex ante belief of sophisticated investors for opportunistic CRA giving G rating if it receives good signal. Thus, the willingness to pay of sophisticated investors is v in period 1, $\omega_2^{S,D}(y), \omega_2^{F,D}(y)$ and $\omega_2^{N,D}(y)$ in period 2. We have:

$$\pi^D(B \mid g; y) = 0 + \delta \max\left\{\beta v, \omega_2^{N,D}(y)\right\};$$
$$\pi^D(G \mid g; y) = v + \delta\left\{[1-(1-\mu)p]\max\left\{\beta v, \omega_2^{S,D}(y)\right\} + (1-\mu)p(1-\beta)\omega_2^{F,D}(y)\right\}.$$

In the second formula, the CRA's payoff in period 1 is independent of y. That is because a G rating always indicates good signal in (semi-)deflating case. And note that $\omega_2^{N,D}(0) = k, \omega_2^{S,D}(0) = \omega_2^{F,D}(0) = v$ (rating deflation), and $\omega_2^{N,D}(1) = \omega_2^{S,D}(1) = \omega_2^{F,D}(1) = w$ (truthful rating).

Similar to the proof of Lemma 4, we use the continuity, the strict monotonicity and the crossing property to prove the uniqueness and existence of $y^*(\delta)$. Thus, semi-deflation $(y, z) = (1, y^*(\delta))$ is a PBE strategy.■

Proof. (Proposition 2). (1) We show that $\delta_h^I(\theta_h, p_h) > \delta_l^I(\theta_l, p_l)$, which implies that the equilibrium condition for rating inflation is more likely to be satisfied in state h than in state l.

First, since $\frac{\partial \bar{v}_h}{\partial \theta_h} > 0$, $\frac{\partial \bar{v}_l}{\partial \theta_l} < 0$, we have

$$\frac{\partial \delta_h^I(\theta_h, p_h)}{\partial \theta_h} < 0 \quad \text{and} \quad \frac{\partial \delta_l^I(\theta_l, p_l)}{\partial \theta_l} > 0.$$

It follows that $\delta_h^I(\theta_h, p_h) \geq \delta_h^I(1, p_h)$ and $\delta_l^I(1, p_l) \geq \delta_l^I(\theta_l, p_l)$.

Next, since $p_h < p_l$, $\frac{w_h}{v_h} > \frac{w_l}{v_l}$, we have

$$\delta_h^I(1, p_h) = \frac{\max\{\beta v_h, w_h\}}{v_h - (1-\mu p_h)\beta v_h} > \frac{\max\{\beta v_l, w_l\}}{v_l - (1-\mu p_l)\beta v_l} = \delta_l^I(1, p_l).$$

Therefore

$$\delta_h^I(\theta_h, p_h) \geq \delta_h^I(1, p_h) > \delta_l^I(1, p_l) \geq \delta_l^I(\theta_l, p_l),$$

or $\delta_h^I(\theta_h, p_h) > \delta_l^I(\theta_l, p_l)$. (2) We show that there exists $\bar{\beta} \in (0,1)$, such that for $\beta > \bar{\beta}$, $\delta_h^D(\theta_h, p_h) > \delta_l^D(\theta_l, p_l)$. That is, the equilibrium condition for rating deflation is more likely to be satisfied in state l than in state h.

First, suppose that $\max\{\beta \bar{v}_s, \bar{k}_s\} = \beta \bar{v}_s$. Then,

$$\delta_s^D = \frac{v_s}{\beta - 1 + (1-\mu)p_s\beta}\frac{1}{\bar{v}_s}.$$

Since $\frac{\partial \bar{v}_h}{\partial \theta_h} > 0$ and $\frac{\partial \bar{v}_l}{\partial \theta_l} < 0$, we have $\frac{\partial \delta_h^D(\theta_h, p_h)}{\partial \theta_h} < 0$ and $\frac{\partial \delta_l^D(\theta_l, p_l)}{\partial \theta_l} > 0$.

Second, suppose that $\max\{\beta \bar{v}_s, \bar{k}_s\} = \bar{k}_s$. Then, $\delta_s^D = \frac{v_s}{\bar{k}_s - \bar{v}_s + (1-\mu)p_s\beta\bar{v}_s}$. Since

$$\begin{aligned}\frac{\partial[\bar{k}_s - \bar{v}_s + (1-\mu)p_s\beta\bar{v}_s]}{\partial \theta_s} &= (k_s - k_{-s}) - [1-(1-\mu)p_s\beta](v_s - v_{-s}) \\ &= R(p_{-s} - p_s)\left\{\left(1 - \alpha_2^{N,D}\right) - [1-(1-\mu)p_s\beta](1-\mu)\right\}\end{aligned}$$

and

$$(1-\alpha_2^{N,D}) - [1-(1-\mu)p_s\beta](1-\mu) > (1-\alpha_2^{N,D}) - (1-\mu) = \mu - \alpha_2^{N,D} > 0,$$

we have $\frac{\partial \delta_h^D(\theta_h, p_h)}{\partial \theta_h} < 0$ and $\frac{\partial \delta_l^D(\theta_l, p_l)}{\partial \theta_l} > 0$. It follows that $\delta_h^D(\theta_h, p_h) \geq \delta_h^D(1, p_h)$ and $\delta_l^D(1, p_l) \geq \delta_l^D(\theta_l, p_l)$.

Third, we show that there exists $\bar{\beta} \in (0,1)$ such that for $\beta > \bar{\beta}$, $\delta_h^D(1, p_h) > \delta_l^D(1, p_l)$. If $\max\{\beta v_s, k_s\} = \beta v_s$, or $\beta \geq \frac{k_h}{v_h}$, then

$$\begin{aligned}\delta_h^D(1, p_h) &= \frac{v_h}{\beta v_h - v_h + (1-\mu)p_h\beta v_h} = \frac{1}{\beta - 1 + (1-\mu)p_h\beta} \\ &> \frac{1}{\beta - 1 + (1-\mu)p_l\beta} = \delta_l^D(1, p_l).\end{aligned}$$

Define $\bar{\beta} \equiv \max\{\frac{k_h}{v_h}, \hat{\beta}_h\}$. Then there exists $0 < \bar{\beta} < 1$ such that for $\beta \geq \bar{\beta}$, $\delta_h^D(\theta_h, p_h) > \delta_l^D(\theta_l, p_l)$.■

Proof. (Corollary 3). (i) The probability of a recession in the second period is either θ_l or $1-\theta_h$. In the Proof of Proposition 2, we've already shown that $\frac{\partial \delta_h^I(\theta_h, p_h)}{\partial \theta_h} < 0$ and $\frac{\partial \delta_l^I(\theta_l, p_l)}{\partial \theta_l} > 0$. It is then straightforward that $\frac{d\delta_h^I(\theta_h)}{d(1-\theta_h)} > 0$. $\frac{\partial \delta_l^I(\theta_l, p_l)}{\partial \theta_l} > 0$ and $\frac{d\delta_h^I(\theta_h)}{d(1-\theta_h)} > 0$ imply that the equilibrium condition for rating inflation ($\delta_s \leq \delta_s^I$) is more likely to be satisfied with a larger recession probability in the second period. (ii) The probability of a boom in the second period is either θ_h or $1-\theta_l$. Again, by the Proof of Proposition 2, we have $\frac{d\delta_h^D(\theta_h)}{d\theta_h} < 0$ and $\frac{d\delta_l^D(\theta_l)}{\partial(1-\theta_l)} < 0$, which mean that the equilibrium condition for rating deflation ($\delta_s \geq \delta_s^D$) is more likely to be satisfied with a larger booming probability in the second period. Hence, we proved the Corollary.■

A.3. Duopoly rating agencies

In this part, we consider the case when there are two opportunistic rating agencies, C_1 and C_2, competing in selling ratings to issuers. Each CRA receives the same but imperfect signal and decides to report {G, B}. The naive investors will punish the CRA when its G-rating security fails or when its B-rating security succeeds.[13] The game rule is as follows:

1. At the beginning of period 1, the issuer approaches both CRAs for ratings for its security.
2. Each CRA posts its rating fee ϕ_i, $i \in \{C_1, C_2\}$, then receives the same private signal, and makes a rating of $m = G$ or $m = B$.
3. The issuer receives the ratings and decides whether to buy and publish one, both, or neither rating. It then sets a price for the security if it is issued. If the security is not issued, there is zero payoff to all parties in period 1.
4. If the security is issued, investors observe its price and rating, and each decides whether to purchase one unit or not. At the end of period 1, the investment outcome, success or failure, is realized.
5. The game then moves to the beginning of period 2, whether or not a security is issued in period 1. In the second period, the game in the first period is repeated.

Everything else of the setting is as before. For exposition, we define:

$$v^{GG} = R\left(1 - \frac{(1-\mu)^2}{(1-\mu)^2 + \mu^2} p\right) - r;$$
$$v^G = R[1 - (1-\mu)p] - r = v.$$

v^{GG} represents the security's value to the sophisticated investors when both CRAs report G truthfully. It also represents the security's value to the naive investors when both CRAs report G, whether truthfully or not. The security's value is v^G to the naive investors if those investors find out that one G-rating is published without seeing the other CRA's report (or the other CRA has been caught lying and thus its rating is disregarded). However, when the sophisticated investors see only one G-rating, they will not buy the security since they infer that the other rating is B, and thus the ratings do not bring any new information. Notice that $v^{GG} > v^G > 0$, that is, two G-ratings provide more information thus more valuable than one G-rating.

To facilitate our analysis, we make the following assumptions:

Assumption 1. $v^G \geq v^{GG} - v^G$.

This is the standard assumption of decreasing returns of G rating, or the value of the first G report is larger than that of the second.

Assumption 2. $v^{GG} > [2-(1-\mu)p]v^G$.

This assumption requires v^{GG} is not too small. It guarantees the existence of both the truthful-rating and the rating-deflation equilibria. Notice this condition is more likely to be satisfied when the signal is less accurate or the bad security is more likely to default.

In the following part, we will first calculate CRAs' profit under different strategies, and then characterize the equilibria. Note that the sophisticated investors will never buy any security in period 2.

As before, we discuss three different regimes: truthful rating, where both CRAs report true signals; rating inflation, where both CRAs inflate ratings; and rating deflation, where both CRAs deflate ratings.

[13] Unlike the monolistic CRA case in Section 2, this setting partially introduces the punishment to CRAs' deflation strategy.

1. **Truthful rating**. Suppose C_2 reports truthfully in period1 . By truthful rating, C_1 can achieve profit:

$$\pi_{C_1}^T(G \mid g; G) = \phi^C + \delta[1-(1-\mu)p]\beta\phi^C;$$
$$\pi_{C_1}^T(B \mid b; B) = 0 + \delta\beta\phi^C,$$

where ϕ^C denotes CRAs' rating fees when both CRAs give the G ratings. Given it is the truthful rating in equilibrium, both the sophisticated and the naive investors will purchase the G-rated security in period 1. However, by deviating to either inflate or deflate, C_1 will obtain respective profit:

$$\pi_{C_1}^T(G \mid b; B) = \beta v^G + \delta(1-\mu p)\beta v^G;$$
$$\pi_{C_1}^T(B \mid g; G) = 0 + \delta(1-\mu)p\beta v^G.$$

By deviating to inflation, C_1 can attract naive investors in period 1, and also receives rating fee in period 2 from naive investors if the security did not default in period 1. By deviating to deflation, C_1 can only obtain rating fee if the security fails. If the CRAs deviate collectively to either deflation or inflation, C_1 will achieve respective profit:

$$\pi_{C_1}^T(B \mid g; B) = \delta\beta\phi^C;$$
$$\pi_{C_1}^T(G \mid b; G) = \phi^C + \delta(1-\mu p)\beta\phi^C.$$

By collectively deviating to deflation, both CRAs can only receive rating fee in period 2 from naive investors. By collectively deviating to inflation, both CRAs can charge both type investors in period 1, but only naive investors in period 2.

2. **Rating inflation**. Suppose C_2 inflates the rating. By inflating or truthful reporting, C_1 can achieve respective profit:

$$\pi_{C_1}^I(G \mid b; G) = \beta\phi^C + \delta(1-\mu p)\beta\phi^C;$$
$$\pi_{C_1}^I(B \mid b; G) = 0 + \delta\mu p\beta v^G.$$

Notice sophisticated investors will not make any investment when both CRAs inflate rating. If both CRAs deviate to truthful report, C_1 will achieve profit:

$$\pi_{C_1}^I(B \mid b; B) = \delta\beta\phi^C.$$

3. **Rating deflation**. Suppose C_2 deflates the rating. By deflating or truthful reporting, C_1 can achieve profit:

$$\pi_{C_1}^D(B \mid g; B) = 0 + \delta\beta\phi^C;$$
$$\pi_{C_1}^D(G \mid g; B) = v^G + \delta[1-(1-\mu)p]\beta v^G.$$

Since G rating indicates good signal under rating deflation regime, sophisticated investors as well as naive investors, will pay v^G for G-rated investment in period 1. If both CRAs deviate to truthful reporting, C_1 will achieve profit:

$$\pi_{C_1}^D(G \mid g; G) = \phi^C + \delta[1-(1-\mu)p]\beta\phi^C.$$

In the above expressions, we have the unknown ϕ^C, which we will try to figure out now. We denote ϕ_{ij} to be CRA i's rating fee in j period, $i \in \{C_1, C_2\}$, and $j \in \{1, 2\}$. Obviously, the issuer prefers two G-ratings to one if $v^G - \phi_{ij} \le v^{GG} - \phi_{C_1j} - \phi_{C_2j}$, or $\phi_{ij} \le v^{GG} - v^G$ given $\phi_{C_1j} = \phi_{C_2j}$ by symmetry. If both CRAs charge $\phi_{ij} = v^{GG} - v^G$, the issuer prefers two G-rating if and only if $v^{GG} - 2(v^{GG} - v^G) \ge 0$, which is exactly Assumption 1.

It is easy to verify that neither CRA has incentive to unilaterally deviate by increasing or decreasing the rating fee $\phi_{ij} = v^{GG} - v^G$. Increasing one's rating fee leads the issuer to purchase the other one's rating which makes the deviation unprofitable; decreasing one's rating fee does not make any difference but reducing its own profit. Furthermore, there is also no room for both CRAs to deviate collectively. Suppose that both CRAs charge higher rating fee. Then the issuer only wants to purchase one G report. In this case, both CRAs have incentive to low its price until reaching $v^{GG} - v^G$ as in Bertrand competition. Suppose that both CRAs charge lower rating fee. This will only reduce their own profit without making any difference. Therefore, we establish $\phi_{ij} = \phi^C = v^{GG} - v^G$, and the issuer's profit is $2v^G - v^{GG}$ and security's price is v^{GG} when both CRAs give G ratings.

By comparing the CRA's profits with both the unilateral and the collective deviations under different regimes, we can obtain the following result:

Proposition 3. *With duopoly CRAs and Assumptions 1 and 2, there exists* $\delta_d^I < \underline{\delta}_d^T \le \bar{\delta}_d^T < \delta_d^D$, *such that, in equilibrium, both CRAs will inflate rating if and only if* $\delta \in (0, \delta_d^I]$; *both CRAs will rate truthfully if and only if* $\delta \in [\underline{\delta}_d^T, \bar{\delta}_d^T]$; *both CRAs will deflate rating if and only if* $\delta \in [\delta_d^D, +\infty)$.

Proof.

(1) **Truthful rating**.
In duopoly case, both CRAs will rate truthfully if and only if

$$\pi^T_{C_1}(G \mid g; G) \geq \max\left\{\pi^T_{C_1}(B \mid g; G), \pi^T_{C_1}(B \mid g; B)\right\};$$
$$\pi^T_{C_1}(B \mid b; B) \geq \max\left\{\pi^T_{C_1}(G \mid b; B), \pi^T_{C_1}(G \mid b; G)\right\}.$$

Therefore, we obtain

$$\underline{\delta}^T_d = \max\left\{\frac{1}{\mu p\beta}, \frac{1}{\frac{v^{GG}-v^G}{v^G} - (1-\mu p)}\right\} \leq \frac{1}{(1-\mu)p\beta} = \bar{\delta}^T_d.$$

Both CRAs will rate truthfully if and only if $\delta \in \left[\underline{\delta}^T_d, \bar{\delta}^T_d\right]$.

(2) **Rating inflation.**
Similarly, both CRAs will inflate rating if and only if

$$\pi^I_{C_1}(G \mid b; G) \geq \max\left\{\pi^I_{C_1}(B \mid b; G), \pi^I_{C_1}(B \mid b; B)\right\}.$$

Solve it and we get

$$\delta \leq \delta^I_d = \min\left\{\frac{1}{\mu p}, \frac{1}{\frac{v^G}{v^{GG}-v^G}\mu p - 1 + \mu p}\right\}.$$

Notice

$$\delta^I_d \leq \frac{1}{\mu p} < \max\left\{\frac{1}{\mu p\beta}, \frac{1}{\frac{v^{GG}-v^G}{v^G} - (1-\mu p)}\right\} = \underline{\delta}^T_d.$$

Therefore, both CRAs will inflate rating if and only if $\delta \in (0, \delta^I_d]$.

(3) **Rating deflation.**
In this case, both CRAs will deflate rating if and only if

$$\pi^D_{C_1}(B \mid g; B) \geq \max\left\{\pi^D_{C_1}(G \mid g; B), \pi^D_{C_1}(G \mid g; G)\right\}.$$

Therefore, we have

$$\delta \geq \delta^D_d = \frac{1}{\left[\frac{v^{GG}-v^G}{v^G} - 1 + (1-\mu)p\right]\beta}.$$

Notice

$$\delta^D_d = \frac{1}{\left[\frac{v^{GG}-v^G}{v^G} - 1 + (1-\mu)p\right]\beta} > \frac{1}{(1-\mu)p\beta} = \bar{\delta}^T_d.$$

Therefore, both CRAs will deflate rating if and only if $\delta \in [\delta^D_d, +\infty)$.■

Proposition 3 demonstrates that when duopoly CRAs compete in selling ratings, just like the monopolistic CRA case, the three rating regimes, rating inflation, truthful rating, and rating deflation, still exist. The intuition behind is still the trade-off between the current rating fees and the future reputation cost. Rating deflation can still survive, even after we introduce the punishment by the naive investors when a B-rating security succeeds, as long as the reputation cost is sufficiently large.

A.3.1. Monopoly versus duopoly

In the monopoly case, we have assumed that sophisticated investors believe the ex ante probability of a CRA to be honest is η. To make it comparable with the duopoly case, we let $\eta = 0$ in this section. Therefore, we have the following relationships with regard to those thresholds in Propositions 1 and 3:

$$\delta^I \geq \delta_d^I; \underline{\delta}^T \leq \underline{\delta}_d^T; \bar{\delta}^T = \bar{\delta}_d^T; \delta^D \leq \delta_d^D.$$

Then we can obtain the following result:

Corollary 4. *Compared to the monopoly case, truthful rating is less likely in the duopoly case.*

Proof. It is easy to verify that

$$\begin{aligned}
\delta^I &= \frac{1}{\mu p} \geq \min\left\{\frac{1}{\mu p}, \frac{1}{\frac{v^G}{v^{GG}-v^G}\mu p - 1 + \mu p}\right\} = \delta_d^I;\\
\underline{\delta}^T &= \frac{1}{\mu p\beta} \leq \max\left\{\frac{1}{\mu p\beta}, \frac{1}{\frac{v^{GG}-v^G}{v^G} - (1-\mu p)}\right\} = \underline{\delta}_d^T;\\
\bar{\delta}^T &= \frac{1}{(1-\mu)p\beta} = \bar{\delta}_d^T;\\
\delta^D &= \frac{1}{(1-\mu)p\beta} \leq \frac{1}{\left[\frac{v^{GG}-v^G}{v^G} - 1 + (1-\mu)p\right]\beta} = \delta_d^D,
\end{aligned}$$

which means $(0, \delta_d^I] \subset (0, \delta^I]$, $\left[\underline{\delta}_d^T, \bar{\delta}_d^T\right] \subset \left[\underline{\delta}^T, \bar{\delta}^T\right]$ and $[\delta_d^D, +\infty) \subset [\delta^D, +\infty)$. Hence, we obtain the result.■

Corollary 4 indicates that competition can exacerbate the information problem and make both CRAs less likely to rate truthfully. The intuition is as follows: If both CRAs always give the same rating (even when they are deviating), the parameter range for δ in the duopoly case will be exactly the same as that in the monopoly case. However, because of the decreasing return of G rating (Assumption 1), a CRA obtains more profit if it gives the only G rating. In other words, a CRA may achieve more profit by the single deviation. To guarantee no single deviation, we need more restrictive condition, which leads to the parameter range for δ shrinking, compared to the monopoly case. This is also true for both the inflation and the deflation regimes.

References

Ashcraft, A., Goldsmith-Pinkham, P., Vickery, J., 2010. MBS ratings and the mortgage credit boom. Working Paper. Federal Reserve Bank of New York.,

Auh, J., 2013. Procyclical Credit Rating Policy. Columbia University.

Bae, K., Kang, J., Wang, J., 2010. Credit Rating Inflation or Deflation? Tests of Two Competing Views on Credit Rating Standard Changes. Working Paper. York University.

Bar-Isaac, H., Shapiro, J., 2011. Credit ratings accuracy and analyst incentives. Am. Econ. Rev. Pap. Proc. 101, 120–124.

Bar-Isaac, H., Shapiro, J., 2013. Rating quality over the business cycle. J. Financ. Econ. 108, 62–78.

Barberis, N., Thaler, R., 2003. A survey of behavioral finance. In: Constantinides, G., Harris, M., Stulz, R. (Eds.), Handbook of the Economics of Finance. Elsevier Science Ltd., North-Holland, Amsterdam, pp. 1053–1123.

Benmelech, E., Dlugosz, J., 2009. The credit ratings crisis. . pp. 161–207.

Bolton, P., Freixas, X., Shapiro, J., 2012. The credit ratings game. J. Financ. 67, 85–112.

Boot, A., Milbourn, T., Schmeits, A., 2006. Credit ratings as coordination mechanisms. Rev. Financ. Stud. 19, 81–118.

Cohn, J., Rajan, U., Strobl, G., 2013. Credit Ratings: Strategic Issuer Disclosure and Optimal Screening. Working Paper. University of Michigan.

Cripps, M., Mailath, G., Samuelson, L., 2004. Imperfect monitoring and impermanent reputations. Econometrica 72, 407–432.

Ferri, G., Liu, L.G., Stiglitz, J., 1999. The procyclical role of rating agencies: evidence from the East Asian crisis. Econ. Notes 28,

Frenkel, S., 2015. Repeated interaction and rating inflation: a model of double reputation. Am. Econ. J. Microecon. 7, 250–280.

Fulghieri, P., Strobl, G., Xia, H., 2014. The economics of solicited and unsolicited credit ratings,. Rev. Financ. Stud. 27, 484–518.

Goel, A., Thakor, A., 2011. Credit Ratings and Litigation Risk. Working Paper. Washington University, St. Louis.

Hirshleifer, D., 2001. Investor psychology and asset pricing. J. Financ. 64 (4), 1533–1597.

Laffont, J., Tirole, J., 1993. A Theory of Incentives in Procurement and Regulation. MIT Press, Cambridge.

Liu, Q., 2011. Information acquisition and reputation dynamics. Rev. Econ. Stud. 78, 1400–1425.

Malmendiera, U., Shanthikumar, D., 2007. Are small investors naive about incentives? J. Financ. Econ. 85, 457–489.

Manso, G., 2013. Feedback effect of credit ratings. J. Financ. Econ. 109, 535–548.

Mariano, B., 2012. Market power and reputational concerns in the ratings industry. J. Bank. Financ. 36, 1616–1626.

Mathis, J., McAndrews, J., Rochet, J., 2009. Rating the raters: are reputation concerns powerful enough to discipline rating agencies? J. Monet. Econ. 56, 657–674.

Opp, C., Opp, M., Harris, M., 2013. Rating agencies in the face of regulation. J. Financ. Econ. 108, 46–61.

Skreta, V., Veldkamp, L., 2009. Ratings shopping and asset complexity: a theory of ratings inflation. J. Monet. Econ. 56, 678–695.